Andreas Tersenov

PHD STUDENT · UNIVERSITY OF CRETE PHYSICS DEPARTEMENT

(+30) 694-577-5095 | atersenov@physics.uoc.gr | https://AndreasTersenov.github.io | AndreasTersenov

Education

University of Crete *Heraklion, Greece*

PHD GRADUATE DEGREE IN PHYSICS *Aug. 2023 - expected Aug. 2026*

- **Project**: ”TITAN - Artificial Intelligence in Astrophysics”, EU ERA Chair
- **Supervisors**: Dr. J.L. Starck, Dr. M. Kilbinger, Prof. V. Pavlidou
- **Topic**: ”Weak Lensing Mass Mapping and Higher-Order Statistics for Precision Cosmology”

University of Crete *Heraklion, Greece*

MSC GRADUATE DEGREE IN PHYSICS *Sept. 2022 - June 2023*

- **Program**: ”Advanced Physics”, specialization: ”Astrophysics and Space Physics”
- **Supervisors**: Dr. J.L. Starck, Prof. V. Pavlidou
- **Thesis**: ”Comparison of weak-lensing mass-mapping techniques. Application to the UNIONS galaxy survey.”

University of Crete *Heraklion, Greece*

BSC UNDERGRADUATE DEGREE IN PHYSICS *2018 - 2022*

- **Grade**: 9.15/10 (Excellent, Top 2% of his class)
- **Supervisor**: Prof. V. Pavlidou, Prof. E. Kiritsis
- **Thesis**: ”Cosmic-ray air-shower simulations: combining mixed galactic composition with new physics above 50 TeV”

Publications

SUBMITTED/PUBLISHED

Impact of mass mapping algorithms on cosmology inference

A. Tersenov, L. Baumont, M. Kilbinger, J.L. Starck
published in Astronomy & Astrophysics 698 (2025): A25. doi, arXiv:2501.06961v2

A plug-and-play approach with fast uncertainty quantification for weak lensing mass mapping

H. Leterme, A. Tersenov, J. Fadili & J.L. Starck
accepted for publication at Astronomy & Astrophysics, arXiv:2603.22006

Euclid preparation: Towards a DR1 application of higher-order weak lensing statistics

Euclid HOWLS Collaboration: S. Vinciguerra, ..., A. Tersenov, et al.
published in Astronomy & Astrophysics 707 (2026): A235. doi, arXiv:2510.04953

Mitigating Baryonic Effects in Weak Lensing with Higher-Order Statistics

A. Tersenov, S. Guerrini, J.L. Starck & M. Kilbinger
submitted for publication to Astronomy & Astrophysics

IN PREPARATION

Validating Wavelet ℓ_1-Norm Theory Predictions Through Cosmological Parameter Inference

A. Tersenov, V. Tinnaneri Sreekanth, M. Kilbinger & J.L. Starck

CONFERENCE PROCEEDINGS

Cosmic-ray air-shower simulations across the ankle: Combining mixed Galactic composition with New Physics above 50 TeV

S. Romanopoulos, A. Tersenov, V. Pavlidou
published in 38th International Cosmic Ray Conference, 2024 (p. 495).

Research Experience

Euclid collaboration - *Lead Contributor & Working Group Member*

GROUP: WEAK LENSING SCIENCE WORKING GROUP *Nov. 2023 - Present*

Description: Serving as an active member of the *Euclid* consortium with specific responsibilities in weak lensing data analysis, pipeline construction, and scientific leadership.

- Lead of the Euclid key paper on "Cosmic shear higher-order statistics analysis of DR1 data with theory", within the Higher-Order Weak Lensing Statistics (HOWLS) project.
- Active contributor to the HOWLS working group, developing and testing pipelines on cosmology inference with higher-order statistics. Lead of the "Mass Mapping and Tomography" brick.
- Active member of the Organization Unit for Level 3 data processing (OU-LE3), contributing to the mass mapping brick.

Institutes of Astrophysics & Computer Science at FORTH, CEA Paris-Saclay - *PhD Fellow* *Heraklion, Greece*

SUPERVISORS: DR. J.L. STARCK, DR. M. KILBINGER, PROF. V. PAVLIDOU *Aug. 2023 - Present*

Description: Developing data-driven methods for weak-lensing mass mapping and cosmological inference, combining physical forward models with ML and simulation-based inference (SBI). Member of the Euclid and UNIONS collaborations.

CosmoStat Laboratory, CEA Paris-Saclay - *Research Intern* *Paris, France*

SUPERVISORS: DR. J.L. STARCK, DR. M. KILBINGER *Feb. 2023 - June 2023*

Description: Analyzed and compared convergence map-making algorithms. Refined the `shear-pipe-peaks` inference pipeline and applied methods to preliminary UNIONS data.

Institute of Astrophysics at FORTH - *Undergraduate/Graduate Research Assistant* *Heraklion, Greece*

SUPERVISOR: PROF. V. PAVLIDOU *Nov. 2020 - June 2023*

Description: Modeled mixed-composition ultra-high-energy cosmic-ray air showers as part of the CIRCE project. Conducted Monte Carlo simulations to study the Galactic–extragalactic transition and test phenomenological new-physics models above 50 TeV.

Crete Center for Theoretical Physics - *Undergraduate/Graduate Research Assistant* *Heraklion, Greece*

SUPERVISOR: PROF. E. KIRITSIS *Oct. 2021 - Oct. 2022*

Description: Studied holographic RG flows in Einstein–Maxwell–dilaton gravity duals of deformed S^3 QFTs. Derived UV/IR asymptotics and analyzed exotic flow solutions.

National University of Science and Technology "MISIS" - Laboratory of Superconducting Metamaterials - *Research Intern* *Moscow, Russia*

SUPERVISOR: DR. ANDREI MALISHEVSKII *Jun. 2019 - Aug. 2019*

Description: Investigated fluxonium qubits coupled to coplanar resonators and electromagnetic waves in quantum metamaterials. Conducted cryogenic measurements and numerical modeling of localized quantum states.

Talks & Posters

TALKS

Fall 2025. ***Impact of mass-mapping and baryonic effects on cosmology inference***. Contributed talk: WL-SWG and OU-SHE Meeting (Marseille, France)

Summer 2025. ***Impact of Weak Lensing Mass Mapping Algorithms on Cosmology Inference***. Talk: COLOURS Workshop (Saclay, France)

Summer 2024. ***Peak counts for UNIONS cosmology***. Contributed talk: UNIONS meeting (Paris, France)

POSTERS

Fall 2025. ***SBI for robust beyond-two-point cosmology***. MINOAS Workshop (Heraklion, Greece)

Fall 2025. ***Impact of baryonic feedback on weak-lensing higher-order statistics*** . Beyond-two-point Statistics Meet Survey Systematics workshop (Kashiwa, Japan)

Spring 2024. ***Impact of mass mapping algorithms on cosmology inference.*** COSMO 21 Conference (Chania, Greece)

Winter 2023. ***Mass mapping and cosmology***. Tonale Winter School on Cosmology (Tonale, Italy)

Open-Source Software (selected)

wl-stats-torch

AUTHOR/MAINTAINER

GPU-accelerated PyTorch library for weak lensing summary statistics (2D & HEALPix).

bar-impact

AUTHOR/MAINTAINER

Collection of tools to analyze the impact of baryons on cosmological weak lensing contours through different summary statistics on HEALPix maps and Implicit-Likelihood Inference.

weaklensing-uq

CO-AUTHOR/CO-MAINTAINER

Repository for the `PnPMass` weak lensing mass mapping algorithm and for the application of distribution-free uncertainty quantification for inverse problems.

cosmostat

CONTRIBUTOR/MAINTAINER

A Python/C++ software package developed at the CosmoStat lab for cosmostatistics.

International Conferences & Schools Organised

Winter School for AstroStatistics — *Sharjah, UAE*

SOC, LECTURER — *Nov 2025*

- International winter school organized by SAASST, bringing together ∼35 international participants.
- Responsibilities: program coordination, applicant selection, preparation and delivery of lectures and tutorials.

Summer School for Astrostatistics in Crete (2025 & 2024 Editions) — *Heraklion, Greece*

ORGANISER, LECTURER — *June 2025 & July 2024*

- Co-organized two editions of an international school, each with ∼35 participants (graduate students and early-career researchers) selected from a highly competitive pool (∼150 applicants in 2025; ∼140 in 2024), from ∼20 countries.
- **Local Organisation & Management** (both editions): Co-led all local logistics, including securing funding, website development, and coordination of venue, catering, merchandise, and social program.
- **Academic Responsibilities** (both editions): Program coordination, applicant selection, and preparation and delivery of lectures and tutorials. The tutorials can be found here.

Euclid-LE3 Meeting — *Hersonissos, Greece*

LOC — *June 2025*

- Supported coordination and logistics for the Euclid Consortium's Level-3 data processing meeting.

COSMO 21 - Statistical Challenges in 21st Century Cosmology — *Chania, Greece*

LOC — *May 2024*

- Assisted in coordination and logistics for a major international cosmology conference with 100 participants.

Teaching Experience

Winter School for AstroStatistics — *Sharjah, UAE*

LECTURER *(Prepared and will deliver the following lectures, and the corresponding hands-on sessions)*: — *Nov 2025*

- Introduction to Bayesian Inference, Advanced Sampling, Simulation-based Inference

Summer School for Astrostatistics in Crete (2025 & 2024 Editions) — *Heraklion, Greece*

LECTURER *(Prepared and delivered the following lectures, and the corresponding hands-on sessions)*: — *June 2025 & July 2024*

- Introduction to Bayesian Inference (2024), Bayesian Model Selection (2024), MCMC & Advanced Sampling (2025), Simulation-based Inference (2024 & 2025)

Introduction to Data Science and Machine Learning (Undergraduate course) — *Heraklion, Greece*

Assistant lecturer — *Fall Semester 2024*

- Delivered lectures on core machine learning concepts and offered supplementary sessions on advanced topics.
- Developed and graded weekly hands-on programming assignments to reinforce theoretical knowledge.
- Mentored students through practical coding and provided guidance on final project work.

Undergraduate Physics Laboratories — *University of Crete, Greece*

Teaching Assistant — *Fall 2022, Spr. 2022, Spr. 2020*

- Guided students through experiments and graded reports for three courses: *Advanced Physics Lab I*, *Physics Lab III – Optics*, and *Physics Lab II – Electricity*.

Honors

Nov. 2025	**2nd Place – NeurIPS 2025 Weak Lensing Uncertainty Challenge** Member of the team that achieved 2nd place in this international competition focused on uncertainty-aware and out-of-distribution detection AI techniques for Weak Gravitational Lensing Cosmology.
2023-2024	**PhD Fellowship** Foundation of Research & Technology - Hellas
2018-2019	**”Chrysanthos and Anastasia Karidis” Bequest Scholarship**

Refereeing

Reviewer for the journal *Astronomy & Astrophysics*.

Research Interests

Computational cosmology; weak gravitational lensing; higher-order statistics; theory modeling; simulation-based inference; generative modeling; deep learning for physical sciences; mass mapping; cosmological survey systematics.

Skills

Programming: Python, C++, MATLAB, JavaScript, HTML, Wolfram Mathematica
Machine Learning & AI: JAX, PyTorch, Pyro/NumPyro, TensorFlow, DeepInverse, Hugging Face, Weights & Biases
High-Performance Computing: Parallel computing (MPI, multiprocessing), GPU acceleration (CUDA), workflow optimization on HPC clusters; experience with multiple national/international HPC facilities
Software Development: Git & GitHub, Docker/Singularity, reproducible pipelines, documentation, continuous integration (CI), testing, and Python packaging
Environments & Tools: Linux/Unix systems, version control, containerized workflows, Jupyter
Human Languages: Greek (native), English (fluent, C2-ECPE), Russian (B2-Saint Petersburg State University)

References

Dr. Jean-Luc Starck — *CEA Paris-Saclay*
Director of Research — *Paris, France*
`jstarck@cea.fr, jlstarck@gmail.com`

Dr. Martin Kilbinger — *CEA Paris-Saclay*
Senior Researcher — *Paris, France*
`Martin.Kilbinger@cea.fr, energie.sombre@gmail.com`

Dr. François Lanusse — *CNRS*
Researcher — *Paris, France*
`francois.lanusse@cnrs.fr`